

EXCLUSIVE

IAS PRELIMS 2026

CURRENT AFFAIRS

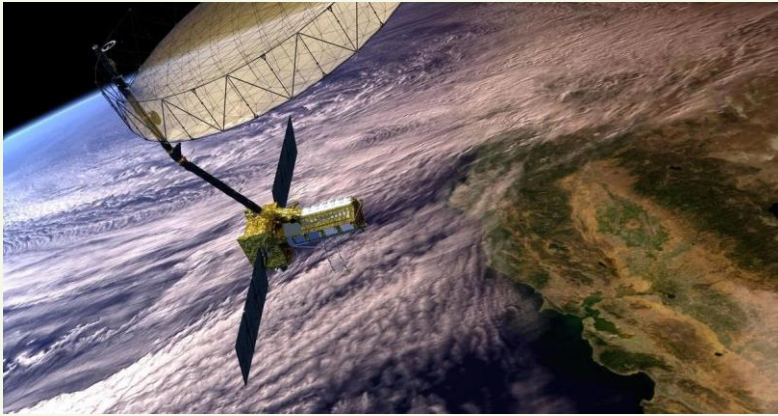
REVISION

GS-1 (DYNAMIC)

MODULE - 100



**SCIENCE
&
TECHNOLOGY- 20**



INDIA LAUNCHES NASA-ISRO EARTH OBSERVATION SATELLITE

- The NASA-ISRO Synthetic Aperture Radar (NISAR) satellite was successfully launched on 30th July, from the Satish Dhawan Space Centre at Sriharikota in Andhra Pradesh.
- The GSLV F16 rocket launched NISAR and was placed in Sun Synchronous Polar Orbit at 747 km altitude.
- The NISAR satellite weighs 2,392 kg. It has a mission life of five years.
- The key applications include shoreline monitoring, storm characterization, mapping of surface water resources, and disaster response.

Watching from above

NISAR, which has a mission life of five years, will observe Earth with a swathe of **242 km** and high spatial resolution



Key applications include: Shoreline monitoring, storm characterisation, mapping of surface water resources, and disaster response

1 It is the first major earth-observing satellite with radars of two frequencies

2 The radars will allow NISAR to monitor both surface and subsurface changes through clouds, smoke, vegetation

■ Its scan-on-receive method will give a spatial resolution of

3-10 metres and centimetre-scale vertical mapping



Giant leap: ISRO's GSLV-F16 carrying the NISAR satellite lifts off from Sriharikota on Wednesday. AP

SCIENCE GOALS OF NISAR

Mission life

5 years

Nominal mission life

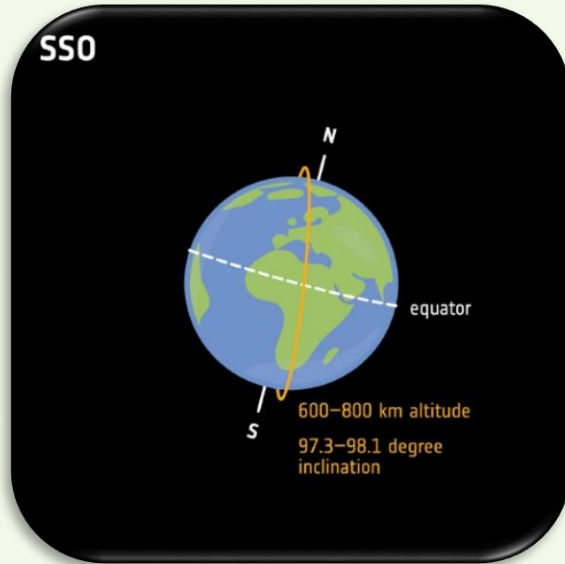
Every 12 days, globally

Science revisit cycle*

NISAR's science goals span the planet

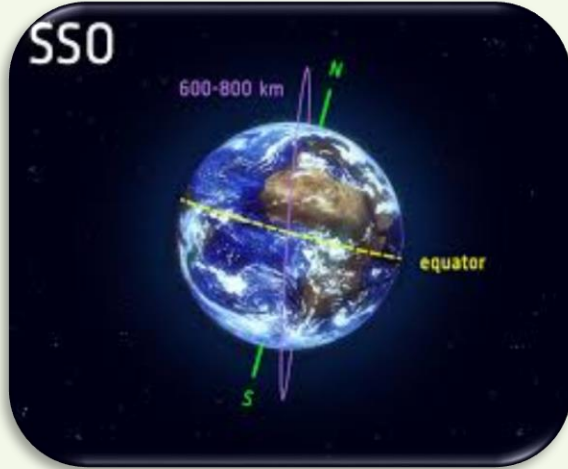
Domain	What it tracks	
Forests		Woody biomass and changes
Agriculture		Extent of active crops
Wetlands		Area and condition
Glaciers		Melting and sea-ice dynamics
Earth movements		Earthquakes, landslides, uplift, aquifers
Oceans		Selected regions with seasonal focus

WHAT DO YOU UNDERSTAND BY A POLAR ORBIT?



- Polar orbits are a type of Low Earth Orbit, typically between 200 to 1,000 km in altitude.
- The Satellites in polar orbits usually travel around Earth from roughly, one pole to the other.
- They do not need to pass exactly over the North and South Poles, a deviation of 10 degrees is still classed as a polar orbit.
- They are particularly useful for global Earth coverage, as satellites orbiting 'up' and 'down' Earth's surface can see every inch of the planet over time as it also rotates below.

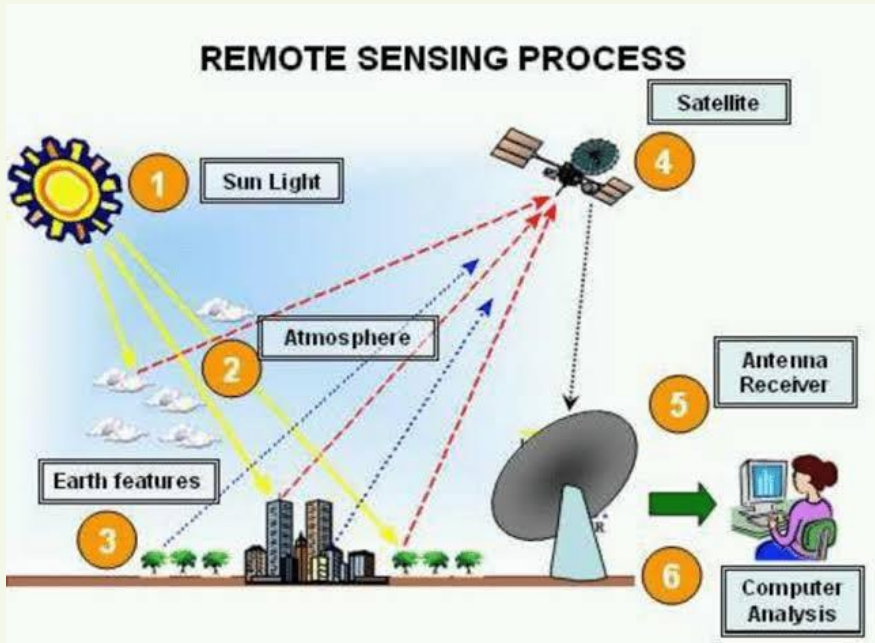
WHAT DO YOU UNDERSTAND BY SUN SYNCHRONOUS ORBIT (SSO)?



- Sun-Synchronous Orbit (SSO) is a particular kind of polar orbit in which satellites are in sync with the Sun.
- Matching Earth's rotation around the Sun, they always appear in the same position relative to our star. The meaning is, they pass over the same spot on Earth.
- This gives facility to compare like for like images overtime to investigate how weather patterns emerge, help predict extreme weather events, monitor emergencies like forest fires or flooding, and to accumulate data on long-term problems like deforestation or rising sea levels.

A Satellite in Sun-Synchronous Orbit usually flies at an altitude of between 600 to 800 km. At 800 km, it will travel at a speed of approximately 7.5 km per second.

WHAT DO YOU UNDERSTAND BY REMOTE SENSING?



- Remote sensing is the acquiring of information from a distance.
- Earth and other planetary bodies are observed via remote sensors or satellites and air crafts that detect and record reflected or emitted energy.
- Remote sensors, which provide a global perspective and a wealth of data about Earth systems, enable data-informed decision making based on the current and future state of our planet.

WE ARE FAMILIAR WITH PASSIVE OPTICAL IMAGES!

Most scientists using remote sensing are familiar with passive, optical images from the U.S. Geological Survey's Landsat, NASA's Moderate Resolution Imaging Spectroradiometer (MODIS), and the European Space Agency's Sentinel-2.

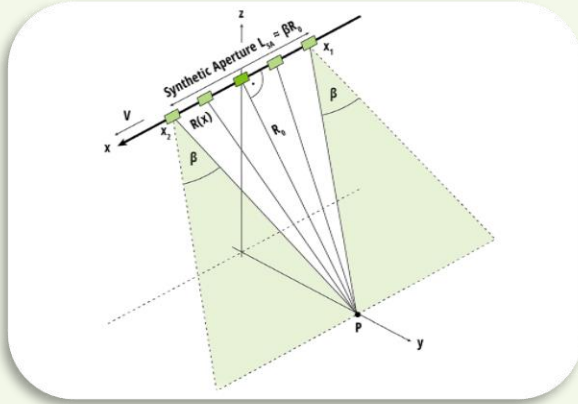


BUT, SAR IS A TYPE OF ACTIVE DATA COLLECTION!

Synthetic Aperture Radar (SAR) is a type of active data collection, where a sensor produces its own energy and then records the amount of that energy reflected back after interacting with the Earth.

While optical imagery is similar to interpreting a photograph, SAR data require a different way of thinking in that the signal is instead responsive to surface characteristics like structure and moisture!

WHAT IS THE MEANING OF TERM “SYNTHETIC” IN SYNTHETIC APERTURE RADAR?



- The spatial resolution of radar data is directly related to the ratio of the sensor wavelength to the length of the sensor's antenna.
- From a satellite in space operating at a wavelength of about 5 cm (C-band radar) - in order to get a spatial resolution of 10 m, you would need a radar antenna about 4,250 m long.
- An antenna of that size is not practical for a satellite sensor in space. Hence, scientists developed the concept - a sequence of acquisitions from a shorter antenna are combined to simulate a much larger antenna, thus providing higher resolution data.

SAR can select frequencies to avoid weather caused signal attenuation. SAR has day and night imaging capability as illumination is provided by the SAR. It is independent of flight altitude and independent of weather.

ISRO and NASA team up for the planet's most advanced Earth observatory

NISAR is the world's first dual-frequency Synthetic Aperture Radar (SAR) mission that can track land deformation, melting glaciers, crop health, and more – all in high resolution, globally, every 12 days.



NISAR (NASA-ISRO Synthetic Aperture Radar) Mission

Launch date July 30, 2025	Launcher GSLV Mk-II
Launch site Satish Dhawan Space Centre, Sriharikota, India	
Data access Free and open for users	
Application areas Glaciers, wetlands, earthquake, biomass, crops, aquifers	

Picture credit: isro.gov.in

NASA-ISRO PARTNERSHIP GOES BACK TO 2014

- NISAR is NASA-ISRO Synthetic Aperture Radar Satellite built by India and USA under a partnership agreement signed in 2014.
- The 2,800kg satellite consists of both L-band and S-band Synthetic Aperture Radar instruments which makes it a dual-frequency image radar satellite.
- While NASA has provided the L-band radar, GPS, a high-capacity solid-state recorder to store data, and a payload data subsystem, ISRO has provided the S-band radar, the Geosynchronous Satellite Launch Vehicle (GSLV) launch system, and spacecraft.
- Another important component is its large 39-foot stationary antenna reflector made of a gold-plated wire mesh. It will be used to focus the radar signals emitted and received by the upward facing feed in the instrument structure.



- It is the first satellite jointly developed by the ISRO and NASA. It is the first satellite to observe the Earth with a dual-frequency Synthetic Aperture Radar (SAR) - NASA's L-band and ISRO's S-band.
- The launch phase has been accomplished with the launch of the GSLV F16 rocket. This is the first time that a GSLV has put a satellite in a Sun synchronous polar orbit.
- The NISAR mission is broadly classified into different phases - launch, deployment, commissioning and science phases. It costs more than \$1.5 billion, making it one of the most expensive and absorbing satellites.

NISAR IS A NEW FRONTIER IN UNDERSTANDING THE EARTH'S DYNAMIC PROCESSES



- Of the satellite's two radars, the L-band (1.2 GHz) made by NASA, can penetrate better through vegetation, ice and so on.
- S-band (3.3 GHz), made by ISRO, gives better resolution. The two bands complement each other, making for a better picture.
- NISAR will not only enhance our understanding of the Earth's dynamic processes but also aid in disaster management and climate research.

The Scientists call the NISAR Mission a new frontier.

THERE ARE SIX APPLICATION GOALS

- Solid earth processes.
- Ecosystems.
- Ice dynamics.
- Coastal and ocean processes.
- Disaster response.
- Additional applications (including tracking groundwater, oil reservoirs and infrastructure like levies, dams, etc.)



The designed lifetime is exactly 5 years. The policy entails that the data NISAR produces will be freely available to all the users within a few hours.

OPERATIONAL DETAILS BY NASA AND ISRO

- While the Mission operations are with the JPL Mission Operations Center, day-to-day flight operations will be led from the ISRO Telemetry Tracking and Command Network in Bengaluru.
- Once NISAR is in orbit, most of its data will be sent through NASA's near-Earth Network facilities in Alaska, Svalbard (Norway), and Punta Arenas (Chile).
- They will be complemented by ISRO's ground stations in Shadnagar and Antarctica. After the raw data arrive, India's national remote sensing centre will process and distribute all products required for Indian users.



LOOK AT THESE STARTLING FIGURES

- As per a report in 2024 by the Centre for Social and Economic Progress (CSEP), India's projected demand for lithium is expected to rise to 20,845 tons by 2047. Nickel demand is projected to increase nearly 30-fold in the same period.
- Rare earths consumption is expected to soar with the growth of wind turbines, EV motors, and advanced electronics.
- Basically, the shift towards renewable energy and e-mobility has put India at the threshold of a resource transformation. However, India is overwhelmingly dependent on imports for these.

THE MINERAL DEMAND IS QUITE HUGE FOR EVs!

**EVs need more minerals than
a conventional vehicle**

	Conventional vehicle	Electric vehicle
Graphite	-	66.3
Nickel	-	39.9
Copper	22.3	53.2
Lithium	-	8.9
Cobalt	-	13.3
Manganese	11.2	24.5



EXPECTED MINERAL DEMAND FOR EVs BY 2030

Expected mineral demand of the EV sector
(in thousand tonnes)

	2020	2030	Growth (%)
Graphite	141.03	2,499.25	1,672
Nickel	80.47	1,566.94	1,847
Copper	110.32	1,632.63	1,380
Lithium	19.83	358.39	1,707
Cobalt	21.12	256.64	1,115
Manganese	25.34	246.28	872

Source: UK, British Geological Survey, IEA, India Exim Bank Research,

FOR LITHIUM, COBALT AND NICKEL ... OUR IMPORT DEPENDENCE IS 100%!

Top producers of rare minerals 2020

Metal	Share of key producers	India's production share
Lithium	Australia (49%), Chile (22%), China (17%), Argentina (8%), Brazil (2%)	-
Graphite	China (65%), Brazil (10%), Madagascar (5%), India (3%), Mozambique (2%)	3%
Cobalt	Congo (68%), Australia (4%), Cuba (4%), Canada (3%), Philippines (3%)	-
Manganese	South Africa (32%), Gabon (16%), China (13%), Australia (10%)	5.32%
Nickel	Indonesia (33%), Philippines (13%), Russia (9%), New Caledonia (8%) Australia (7%)	-



WHAT ARE INDIA'S EFFORTS FOR TRANSITION TOWARDS THESE MINERALS?



- Several measures have taken shape since January 2025, when the Rs. 16,300-crore National Critical Minerals Mission was announced.
- The government owned Khanij Bidesh India Limited (KABIL) has been tasked with acquisitions backed by a Rs. 4,500 crore war chest.
- Amendments to the Mines and Minerals (Development and Regulation) Act now allow the use of funds from the National Mineral Exploration Trust (formed through licence and mining lease payments and royalties) for overseas strategic acquisitions - potentially yielding another Rs. 8,700 crore.
- Further, Rs. 500 crore has been earmarked for stockpiling critical minerals as a hedge against supply shocks, while Rs. 1,500 crore is being set aside to incentivise recyclers who extract minerals from used equipment.
- The government has also committed to 1,200 exploration projects and the auction of 100 critical mineral blocks by 2031.



WHY ARE WE TALKING ABOUT HIGH FRUCTOSE CORN SYRUP (HFCS) AND CANE SUGAR (CS)?

- Donald Trump, the USA President recently announced that beverage giant Coca-Cola would replace High Fructose Corn Syrup used in its drinks with Cane Sugar in the USA.
- Coca-Cola in USA uses high fructose corn syrup, whereas it uses cane sugar in India as the primary sweetener.
- The main difference between cane sugar and corn syrup lies in their sources and chemical composition.
- Cane sugar is sucrose derived from sugar cane, while corn syrup (high fructose corn syrup) is made from corn starch and contains a mix of glucose and fructose.
- Cane sugar typically costs more than high fructose corn syrup.

LET US LEARN FROM THE BASICS OF STARCHES AND SUGARS!



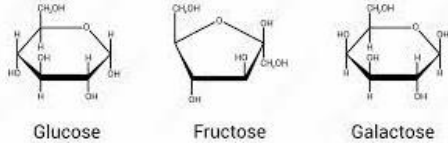
- Starches and sugars are two types of carbohydrates found in food.
- Carbohydrates are the organic compounds made-up of Carbon(C), Hydrogen (H) and Oxygen (O) in the ratio of 1:2:1, for example the sugar glucose has the chemical formula $C_6H_{12}O_6$. Sugar is the monomer unit of complex carbohydrates.
- Autotrophs produce the simple monosaccharide glucose from carbon dioxide and water during photosynthesis.
- Starch is polysaccharides that contain long carbohydrate chains of glucose. Plants store starch in staple foods like fruits, tubers like potatoes., seeds like rice, wheat, corn and cassava.

The main difference between sugar and starch is - the sugars are disaccharides or monosaccharides, whereas starch is a polysaccharide.

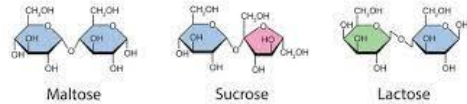
WHAT EXACTLY YOU UNDERSTAND BY SUGARS?

WHAT ARE MONOSACCHARIDES AND DISACCHARIDES?

Monosaccharides



Disaccharides



- Sweet and soluble carbohydrates are generally called sugars. Both monosaccharides and disaccharides are sugars.
- Examples for Monosaccharides are Glucose, Fructose and Galactose.
- Examples for Disaccharides are Lactose, Sucrose and Maltose.

MONOSACCHARIDES

- Glucose is the organic molecule used in the metabolism as the energy source. All other forms of carbohydrates are also converted into glucose during the digestion. It is more commonly found in ripe fruits.
- Fructose is the fruit sugar, it is also found in honey.
- Galactose is present in mammal's milk.

DISACCHARIDES

- These are formed by joining two monosaccharides together.
- Sucrose is found in sugar cane and sugar beets. Common table sugar is an example of sucrose.
- Lactose is found in mammal's milk.
- Maltose is found in cereals like barley. It is the product of starch digestion.

IS CANE SUGAR A HEALTHY ALTERNATIVE IN COMPARISON TO HFCS?



- HFCS does contain slightly more fructose - about 55% compared to 50% in cane sugar. But experts say the difference is not enough to make one a healthier option.
- The American Heart Association recommends no more than 25 grams of added sugar daily for women and 36g for men. But most Americans consume two to three times that.
- Both the sweeteners can contribute to obesity, diabetes and high blood pressure.
- However, Fructose is processed in the liver, where excess amounts are turned into triglycerides - a type of fat linked to heart disease. It also doesn't trigger insulin, the hormone that signals fullness. Overtime, too much fructose can lead to insulin resistance and increase the risk of type-II diabetes.

Added sugar in whatever form, should be limited!

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